

Sentiment Analysis on Product Reviews

Internship Project — Week 1 | Xlofy AI | April–May 2026

Project Overview

This project builds a Sentiment Analysis system for Amazon Fine Food Reviews. Using Python, Pandas, and TextBlob, we loaded 5,000 reviews, cleaned the data, classified each review as Positive, Negative, or Neutral, and visualized the results through three distinct charts.

Dataset & Methodology

Source: Amazon Fine Food Reviews (Kaggle). We retained the first 5,000 rows, then removed 30 null reviews and 48 duplicates, leaving 722 clean reviews for analysis. TextBlob's polarity scoring (range: -1.0 to +1.0) was applied to each review text, and the score was mapped to a sentiment label.

Key Findings

Metric	Value
Total Reviews Analyzed	722
Positive Reviews	490 (67.9%)
Negative Reviews	230 (31.9%)
Neutral Reviews	2 (0.3%)
Average Polarity Score	0.1896

Insights & Summary

After analyzing 722 product reviews, **67.9% were classified as Positive**, confirming that the majority of customers are satisfied with their purchases. Negative reviews — comprising 31.9% — frequently contained words such as *terrible*, *awful*, *nothing*, *very*, *product*, pointing to complaints about product quality and misleading descriptions.

One surprising finding was that TextBlob occasionally labelled 4-star reviews as Negative — this occurs because customers sometimes use cautious or critical language even within a mostly positive review. The average polarity of 0.19 confirms an overall positive customer experience across the dataset.

Business Recommendation

Recommendation: The business should invest in stricter quality control and ensure that product descriptions on the platform accurately reflect the actual item. The most common complaints relate to quality and unmet expectations — addressing these two areas could measurably shift the 32% negative review rate downward, improving overall customer satisfaction and ratings.

Tools used: Python 3, Pandas, TextBlob, Matplotlib, Seaborn | Submitted to Xlofy AI